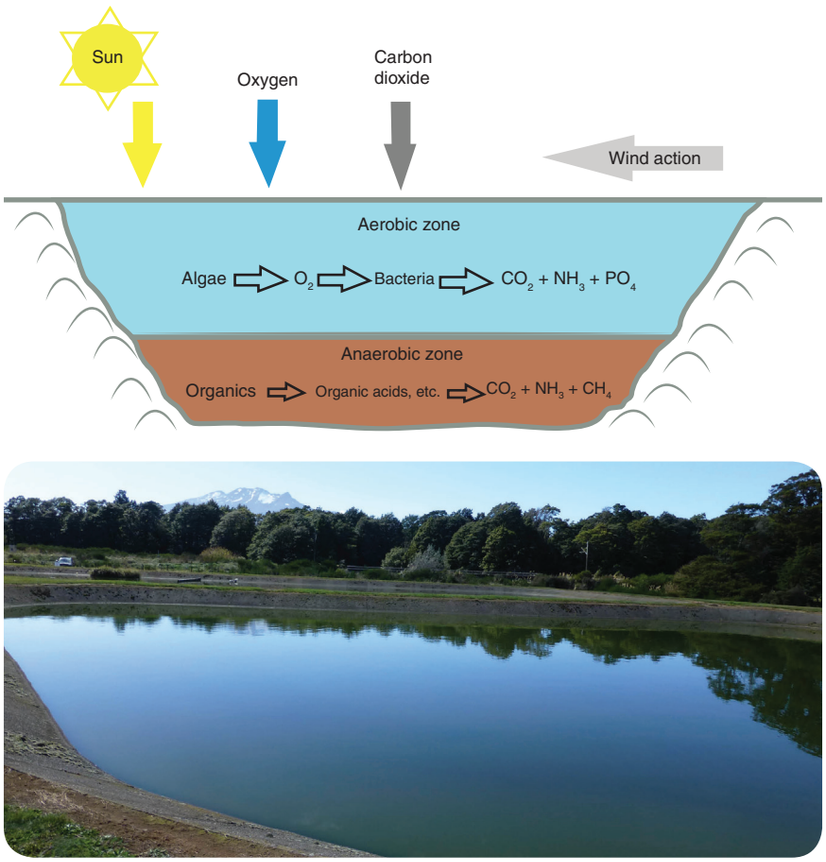


Figure 9.10: Oxidation Ponds

Sludge digestion may also take place aerobically The sludge is vigorously aerated in an open tank for about 20 days. Methane gas is not formed in this process.

Digested sewage sludge is usually dewatered before disposal. Sludge-drying beds provide the simplest method of dewatering. A digested sludge slurry is spread on an open bed of sand and allowed



Septic tank is a small scale anaerobic treatment process. It is commonly used in rural areas. It is simple, inexpensive and satisfactory if properly operated. The septic tank consists of an underground sedimentation container into which sewage from a home enter and is retained for a short time. The organic matter in the sewage settles to the bottom of the tank where it is covered by a thin organic film that excludes oxygen. Anaerobic bacteria in the sediment digests the organic matter into simpler chemical compounds and gases. The gases are then discharged through a vent in the tank. Liquids in the tank rise and overflow through an outlet pipe and are distribute in the surrounding soil. As the water trickles through the soil any remaining organic matter is decomposed by aerobic prokaryotes. Septic tank should not be located near water supplies because not all bacterial pathogens are removed by this treatment. Undigested solids in the bottom of the septic tanks must be periodically removed.

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